

SHOULD-COST ANALYSIS REPORT

Generated: 13 Aug 2026 · 12:43

Bottom-Up Manufacturing Cost Model · Fully Traceable Rate Data

Spur Gear, m3 z38, 30mm Face (test-gear-m3-z38.step)

Commodity: GEAR · Currency: CNY · FX Rate: 9.0498 to GBP · Operations: 5 · Region: CN

Total Should-Cost	Material	Process	Labour	Margin
¥137.21	40.9%	28.0%	7.1%	7.4%

Model Confidence:

Low

· 5 traced operations · 12 data points auditable

Engine Warnings (7)

No straightening allowed for. A quenched gear or shaft normally needs a press straighten to hold runout, and it is the most commonly omitted line in gear heat-treat cost. Confirm it is genuinely not required, or enable it.

No shot peening allowed for. A hardened gear at ISO class 7 or tighter is usually a power-transmission part, where root-fillet peening is standard practice and is often quoted inside the heat-treat package. Confirm whether the supplier includes it.

88 of 88 gear shop parameters are representative values, not verified data (source set: "representative (no plant data supplied)"). Cutting speeds, feeds, tool life and heat-treat rates were not taken from a standard, a machine-builder catalogue or your plant. This estimate shows the right structure and is NOT a quotable number until the shop data is replaced.

Gear route: Gear hobbing -> Chamfer and deburr -> Wash / degrease -> Carburise, quench and temper -> Wash / degrease -> Temper -> Gear grinding -> Gear inspection — cutting cycle 2.54 min/part; perishable tooling £0.59/part; NRE £18500 amortised over 200,000 parts.

Raw Material bucket = blank material £4.76 + heat-treat package £1.31/part (wash / degrease £0.16 + gas carburise + oil quench (sealed-quench batch) £0.84 + wash / degrease £0.16 + temper (stand-alone / double temper) £0.16). Bought by weight, so it carries no machine hours. Blank turning and all tooth cutting/finishing are separate operations in the process bucket.

Heat-treat rate derivation — Gas carburise + oil quench (sealed-quench batch) @ CN: 600 kg/load / 7.0000 h x 8000 h x 0.82 OEE = 562,286 kg/yr. energy 0.0810 + labour 0.0225 + capital 0.0749 + maint 0.0221 + consumables 0.0496 + fixtures 0.0132 + overhead 0.0451 + QC 0.0226 = conversion 0.3310 + scrap 0.0060 = in-house 0.3370 + SG&A 0.0270 + margin 0.0291 + freight 0.0075 = buy 0.4006 GBP/kg

Heat-treat energy floor: £0.081/kg is the energy cost alone for gas carburise + oil quench (sealed-quench batch) in CN. Divide any supplier quote by it — below about 2.5x, the quote is not covering labour, capital and overhead, and the reason is worth asking: cheaper energy than grid tariff, a denser furnace charge than assumed, a loss-leader to win the machining package, or a rate not covering capital.

Geometry Provenance — MEASURED (OCCT B-rep kernel)

Volume, weight and every feature were measured from the CAD solid by the Open CASCADE kernel. Measured volume 266.0 cm³; mass for the selected material family 2.088 kg.

Material cost and geometry-derived machining are grounded in the actual solid — not an estimate.

Key Assumptions

Annual volume: 200,000 · Region: CN · Commodity: gear

Alloy / material: 20MnCr5 (Gear Steel) · Net weight: 2.088 kg (measured)

Material utilisation: 100% · Overhead: 12% of factory base · Margin: 8% of subtotal · Operations: 5

§1 — 8-Bucket Cost Breakdown

Cost Bucket	Amount (CNY)	% of Total	Cost Mix Bar
1. Raw Material	¥56.08	40.9%	
2. Process (Machine)	¥38.41	28.0%	
3. Direct Labour	¥9.81	7.1%	
4. Tooling (amortised)	¥6.22	4.5%	
5. Packaging	¥1.18	0.9%	
6. Logistics	¥2.08	1.5%	
Factory Cost	¥113.78	82.9%	
7. Overhead (SG&A)	¥13.26	9.7%	
Subtotal	¥127.05	92.6%	
8. Supplier Margin	¥10.16	7.4%	
TOTAL SHOULD-COST	¥137.21	100.0%	

§2 — Commercial Parameters

Overhead Rate	12.0%	Supplier Margin	8.0%
Packaging / part	¥1.18	Logistics / part	¥2.08
Total Tooling Cost	¥1244111.85	Amortisation Volume	200,000 parts

§3 — Material Detail

Parameter	Value	Unit	Notes
Material ID	mat-steel-20mncr5	ID	
Grade / Specification	20MnCr5 (Gear Steel)		Index-anchored 2026-07 (see FORGING PRICING BASIS). European case-hardening gear steel — transmission gears, shafts; excellent hardenability and grain control. Regional adj. ×0.880 (millSteel)
Region	China		
Net Finished Weight	2.0880 kg	kg	Weight in finished part
Direct Material Cost	¥43.96	CNY	Bypasses weight-based model
Data Confidence	Medium		
Effective Date	2026-07		

§4 — Operations Detail

4A Machine Operations

Operation	Machine Class	Rate / hr	Cycle (min)	Parts / Cycle	OEE %	Machine Cost
Blank turning (face, OD, bore)	CNC Lathe (2-axis)	¥185.20	4.93	1	80.0%	¥19.04
Gear hobbing	CNC Gear Hobber — small ("d m 4 , "d Ø 2 0 0)	¥265.03	0.95	1	80.0%	¥5.23
Chamfer and deburr	Gear Chamfer / Deburr Cell	¥123.89	0.43	1	80.0%	¥1.10
Gear grinding	Generating Gear Grinder (threaded wheel, "d m 6)	¥603.72	0.88	1	80.0%	¥11.09
Gear inspection	CNC Gear Checker (profile / lead / pitch)	¥272.86	0.34	1	80.0%	¥1.95
TOTAL						¥38.41

4B Labour Detail

Operation	Labour Grade	Rate / hr	Man ning	Lab. min	Eff %	Labour Cost	Op Total
Blank turning (face, OD, bore)	Skilled Machinist	¥71.49	1	4.93	90.0 %	¥6.53	¥25.57
Gear hobbing	Skilled Machinist	¥71.49	1	0.95	90.0 %	¥1.25	¥6.48

Operation	Labour Grade	Rate / hr	Man ning	Lab. min	Eff %	Labour Cost	Op Total
Chamfer and deburr	Semi-skilled Operator	¥49.77	1	0.43	90.0 %	¥0.39	¥1.50
Gear grinding	Skilled Machinist	¥71.49	1	0.88	90.0 %	¥1.17	¥12.26
Gear inspection	CMM / Quality Inspector	¥72.40	1	0.34	90.0 %	¥0.46	¥2.41
TOTAL						¥9.81	¥48.22

4C Machined Features — Geometry Audit

Feature	Ø×depth / area	Qty	Operation	Min	Cost	Costed ?
Hole / bore	Ø40.0 × 30 thru	1	Helical mill / bore	2.00	¥8.48	Yes
Face	8865 mm²	2	Face milling (facing)	2.62	¥11.09	Yes
TOTAL costed		3		4.62	¥19.56	Yes

\$5 — Machine Rate Buildup

CNC Lathe (2-axis) [mach-lathe-cnc] · Confidence: Low · UK Tier-2 CNC turning centre (Doosan/Hyundai class). Target £40/hr. Jun 2026 | Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh

Cost Component	Annual Cost (CNY)	Rate/hr @80% util	Notes
Depreciation	¥199095.02	¥62.22	4,000 hr/yr available
Maintenance	¥109502.26	¥34.22	
Energy	¥55085.58	¥17.21	
Floor Space	¥39819.00	¥12.44	
Indirect Support	¥109502.26	¥34.22	
Finance Cost	¥79638.01	¥24.89	
TOTAL — CNC Lathe (2-axis)	¥592642.14	¥185.20	Effective hours: 3200/yr

CNC Gear Hobber — small ("dm4, "d Ø 200) [gear-hob-small] · Confidence: Low · REPRESENTATIVE · Target ~£56/hr. Replace with plant data. | Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh

Cost Component	Annual Cost (CNY)	Rate/hr @80% util	Notes
Depreciation	¥308597.29	¥96.44	4,000 hr/yr available
Maintenance	¥159276.02	¥49.77	
Energy	¥71611.25	¥22.38	
Floor Space	¥54751.13	¥17.11	
Indirect Support	¥149321.27	¥46.66	
Finance Cost	¥104524.89	¥32.66	
TOTAL — CNC Gear Hobber — small ("dm4, "d Ø 200)	¥848081.84	¥265.03	Effective hours: 3200/yr

Generating Gear Grinder (threaded wheel, "dm6) [gear-grind-generating] · Confidence: Low · REPRESENTATIVE · Target ~£130/hr. Replace with plant data. | Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh

Cost Component	Annual Cost (CNY)	Rate/hr @78% util	Notes
Depreciation	¥821266.97	¥263.23	4,000 hr/yr available
Maintenance	¥338461.54	¥108.48	
Energy	¥151485.34	¥48.55	
Floor Space	¥109502.26	¥35.10	
Indirect Support	¥288687.78	¥92.53	
Finance Cost	¥174208.14	¥55.84	

**TOTAL — Generating Gear
Grinder (threaded wheel, "d m 6)**

¥1883612.04

¥603.72

Effective hours: 3120/yr

Gear Chamfer / Deburr Cell [gear-deburr] · Confidence: Low · REPRESENTATIVE — UK chamfer/deburr cell class. Target ~£26/hr. Replace with plant data. | Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh

Cost Component	Annual Cost (CNY)	Rate/hr @82% util	Notes
Depreciation	¥129411.76	¥39.45	4,000 hr/yr available
Maintenance	¥69683.26	¥21.24	
Energy	¥33051.35	¥10.08	
Floor Space	¥39819.00	¥12.14	
Indirect Support	¥89592.76	¥27.31	
Finance Cost	¥44796.38	¥13.66	
TOTAL — Gear Chamfer / Deburr Cell	¥406354.52	¥123.89	Effective hours: 3280/yr

CNC Gear Checker (profile / lead / pitch) [gear-checker] · Confidence: Low · REPRESENTATIVE — UK CNC gear metrology class (Klingelnberg/Gleason type). Target ~£56/hr. Replace with plant data. | Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh

Cost Component	Annual Cost (CNY)	Rate/hr @65% util	Notes
Depreciation	¥288687.78	¥111.03	4,000 hr/yr available
Maintenance	¥109502.26	¥42.12	
Energy	¥27542.79	¥10.59	
Floor Space	¥59728.51	¥22.97	
Indirect Support	¥149321.27	¥57.43	
Finance Cost	¥74660.63	¥28.72	
TOTAL — CNC Gear Checker (profile / lead / pitch)	¥709443.24	¥272.86	Effective hours: 2600/yr

How to read the machine rate

Rate/hr = (annual depreciation + maintenance + energy + floor + indirect + finance) ÷ effective hours, where effective hours = available hours × utilisation.

Effective hours encode the shift pattern: a lower hours-base raises £/hr and a higher one lowers it. A challenge from a supplier will target this divisor and whether depreciation is machine-only or full-cell (machine + dosing furnace + trim + robots/extraction) — state your basis when defending the number.

§6 — Rate Traceability

Field	Value	Unit	Source / Reference	Rate ID	Conf
-------	-------	------	--------------------	---------	------

Field	Value	Unit	Source / Reference	Rate ID	Conf.
rawMaterial.directCost	4.8571	£	Pre-computed by commodity module	mat-steel-20mncr5	Medium
rawMaterial.consumablesCostPerPart	1.3399	£	Per-part consumable (core/wax/shell)	mat-steel-20mncr5	Medium
Blank turning (face, OD, bore).machineRatePerHr	20.4647	£/hr	UK Tier-2 CNC turning centre (Doosan/Hyundai class). Target £40/hr. Jun 2026 Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh	mach-lathe-cnc	Low
Blank turning (face, OD, bore).labourRatePerHr	7.9000	£/hr	Regional benchmark China — 2026 Q2	lab-uk-skilled	Low
Gear hobbing.machineRatePerHr	29.2853	£/hr	REPRESENTATIVE — UK small CNC hobber class. Target ~£56/hr. Replace with plant data. Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh	gear-hob-small	Low
Gear hobbing.labourRatePerHr	7.9000	£/hr	Regional benchmark China — 2026 Q2	lab-uk-skilled	Low
Chamfer and deburr.machineRatePerHr	13.6897	£/hr	REPRESENTATIVE — UK chamfer/deburr cell class. Target ~£26/hr. Replace with plant data. Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh	gear-deburr	Low
Chamfer and deburr.labourRatePerHr	5.5000	£/hr	Regional benchmark China — 2026 Q2	lab-uk-semiskilled	Low
Gear grinding.machineRatePerHr	66.7113	£/hr	REPRESENTATIVE — UK generating (threaded-wheel) gear grinder class. Target ~£130/hr. Replace with plant data. Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh	gear-grind-generating	Low
Gear grinding.labourRatePerHr	7.9000	£/hr	Regional benchmark China — 2026 Q2	lab-uk-skilled	Low
Gear inspection.machineRatePerHr	30.1513	£/hr	REPRESENTATIVE — UK CNC gear metrology class (Klingelnberg/Gleason type). Target ~£56/hr. Replace with plant data. Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh	gear-checker	Low
Gear inspection.labourRatePerHr	8.0000	£/hr	Regional benchmark China — 2026 Q2	lab-uk-inspector	Low

§7 — Should-Cost with Uncertainty

Monte-Carlo · Low confidence

Estimate (± band)	P10 · optimistic	P50 · median	P90 · conservative	Band	CV
¥137.21 ± 17.4%	¥114.39	¥135.75	¥162.08	moderate	13.7%

Range reflects how well each cost driver is known (High / Medium / Low confidence). Tighten it by attaching CAD, a BOM, or logging an actual quote.

Why confidence is Low

Main band drivers: 10 rate(s) at Low confidence; tooling amortisation carries the widest per-bucket spread.

What's excluded from this unit cost

One-time NRE — tooling / die, fixtures and CNC programming — is amortised separately, not in this unit cost.

Import duty and international freight (regional table is Ex-Works).

Formal embodied-carbon reporting (figures are indicative cradle-to-gate — replace with supplier EPDs).

§8 — Sensitivity Analysis

± 10% driver swing · ranked by ¥ impact

Cost Driver	Base	-10% cost	+10% cost	Range ¥
Direct material cost	¥43.96	-3.9%	+3.9%	¥10.63
Blank turning (face, OD, bore): Cycle Time	0.08hr	-2.3%	+2.3%	¥6.19
Blank turning (face, OD, bore): Machine Rate	¥185.20/hr	-1.7%	+1.7%	¥4.61
Gear grinding: Cycle Time	0.01hr	-1.1%	+1.1%	¥2.97
Overhead %	12%	-1%	+1%	¥2.86
Gear grinding: Machine Rate	¥603.72/hr	-1%	+1%	¥2.68
Blank turning (face, OD, bore): Labour Rate	¥71.49/hr	-0.8%	+0.8%	¥2.17
Gear hobbing: Labour Rate	¥71.49/hr	-0.8%	+0.8%	¥2.17

§9 — Regional Cost Comparison

Ex-Works · per-region should-cost · vs China

Region	Material	Process	Labour	Tooling	Overhead	Ex-Works	Logistics	Total	vs China
United Kingdom (GBP)	¥63.73	¥69.84	¥35.31	¥11.31	¥17.68	¥197.88	¥1.44	¥211.16	+54%
Germany (EUR)	¥65.64	¥73.34	¥57.07	¥11.88	¥22.45	¥230.37	¥1.65	¥245.60	+79%
France (EUR)	¥65.00	¥64.26	¥41.02	¥10.41	¥18.62	¥199.30	¥1.65	¥212.97	+55%
Spain (EUR)	¥63.73	¥61.46	¥25.86	¥9.95	¥15.01	¥176.02	¥1.65	¥188.45	+37%
Poland (PLN)	¥61.82	¥50.29	¥16.05	¥8.14	¥11.37	¥147.67	¥1.72	¥158.54	+16%
Turkey (TRY)	¥57.36	¥41.91	¥8.92	¥6.79	¥8.80	¥123.77	¥1.87	¥133.40	-3%
China (CNY)	¥56.08	¥38.41	¥9.81	¥6.22	¥13.26	¥123.79	¥2.08	¥137.21	Base
India (INR)	¥57.36	¥36.32	¥6.24	¥5.88	¥7.48	¥113.27	¥2.15	¥122.41	-11%
Mexico (MXN)	¥60.54	¥41.91	¥10.34	¥6.79	¥9.15	¥128.73	¥1.94	¥138.60	+1%
United States (USD)	¥63.73	¥59.37	¥46.37	¥9.61	¥16.70	¥195.77	¥1.79	¥209.24	+52%

Per-region should-cost using regional labour, energy and rate benchmarks. Tooling held fixed. Ex-Works — excludes import duty + international freight. Indicative — confirm with RFQ.

§10 — Embodied Carbon

8.85 kgCO2e · 4.24 /kg · cradle-to-gate

Material	4.38 kgCO2e	Material factor	2.10 kg/kg (Steel)
Process	4.13 kgCO2e	Grid intensity	0.550 kg/kWh · 7.52 kWh
Logistics	0.33 kgCO2e	Total	8.85 kgCO2e

Indicative cradle-to-gate factors — replace with supplier EPDs for formal reporting.

[INFO] Split routing — consolidation worth quoting

Finding	5 machining operation(s) across 5 separate stations. Each station is a re-fixturing: fixture cost, handling time, and datum-transfer variation.
Impact	Medium · up to 6% potential saving
Action 1	Quote a multi-axis (4/5-axis) consolidation of the milled and drilled features against the split routing
Action 2	Redesign features (deep pockets, undercuts) that force additional setups

§12 — Cost-Reduction Opportunities (ranked by saving)

7 opportunities · combined ¥16.88/part (~12%)

Ranked on money per part against this costing. The combined figure is the root-sum-square of the top three, capped at 40% — overlapping actions do not save twice, so the column is deliberately not summed.

#	Category	Action	Why it is on the list	Save/p art	%	Timeframe	Risk
1	Process & Automation	Re-quote Machining on the Cost-Optimal Routing	The costed 5-station routing books £6.86/part of machining; the same content on the cheapest capable machines costs £6.18 — a £0.67/part (10%) machine-mix saving. Note: 5-axis consolidation was ranked and does NOT win on this part.	¥13.50	9.8%	Quick Win	Low
2	Process & Automation	Attack the Bottleneck: "Blank turning (face, OD, bore)"	One operation — Blank turning (face, OD, bore) — carries 53% of the conversion cost. Improving anything else first is working on the wrong problem.	¥5.11	3.7%	Medium Term	Medium
3	Process & Automation	OEE Improvement Programme (TPM)	OEE at 80% vs 85% world-class target. Each 5% OEE gain reduces effective machine cost by 5%.	¥2.06	1.5%	Medium Term	Low
4	Design & Geometry	Tolerance and Surface-Finish Relaxation	Grinding/honing/reaming content is 8.9% of the part cost. Each of those operations exists to hit a callout — challenge the callouts before paying for them.	¥8.23	6.0%	Quick Win	Low
5	Design & Geometry	Fix the burr at source (tooling discipline, gate redesign); where finishing must remain, tumble/vibratory beats manual	Manual finishing after the primary process — Deburring/fettling/rework appears as its own operation — a symptom of an upstream condition (tool wear, gating, die state) being paid for by hand, every part.	¥4.12	3.0%	Quick Win	Low
6	Material	Consumables Rationalisation (Cores, Patterns, Shell)	Per-part consumables are 22% of the material line. Core count, pattern life and shell recipe are all design-controllable.	¥5.93	4.3%	Medium Term	Medium
7	Commercial & Sourcing	Raw-Material Price Indexation Clause	With material at 40.9% of cost, the quote embeds the supplier's hedge against metal/resin volatility. An index-linked clause removes that risk premium from the piece price.	¥2.80	2.0%	Quick Win	Low

§13 — Opportunity by Category

4 categories with an actionable lever

Category	Opportunities	Best single action	Best save/part	Category total (indicative)
Process & Automation	3	Re-quote Machining on the Cost-Optimal Routing	¥13.50	¥20.67
Design & Geometry	2	Tolerance and Surface-Finish Relaxation	¥8.23	¥12.35
Material	1	Consumables Rationalisation (Cores, Patterns, Shell)	¥5.93	¥5.93
Commercial & Sourcing	1	Raw-Material Price Indexation Clause	¥2.80	¥2.80

§15 — Implementation Roadmap

Quick Wins — Implement Immediately

- Re-quote Machining on the Cost-Optimal Routing
- Tolerance and Surface-Finish Relaxation
- Raw-Material Price Indexation Clause